
Retrieval Augmented Generation for Financial QA

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Abstract

Large language models (LLMs) often struggle with factual accuracy on queries. This is especially true for financial data which requires recent, domain-specific knowledge. Retrieval-augmented generation (RAG) has emerged as a promising approach for enhancing factual accuracy in question-answering systems by directly sourcing relevant information from external databases. However, financial documents pose additional challenges due to the presence of structured and unstructured data as well as specialized terminology. In this work, we benchmark advanced RAG methods for financial question-answering. Using the FinanceBench dataset, we assess generations on three types of queries: domain-relevant, metric generation and novel-generation. We build a strong baseline model with an accuracy of 37%, and explore 6 other RAG techniques, leading to an accuracy of up to 54%.

1 Introduction

Large Language Models (LLM) have demonstrated impressive text generation and question-answering capabilities, but the issue of hallucination [1], i.e. generating plausible but factually incorrect answers persists. This is particularly evident in queries that demand knowledge the model is not pre-trained on, such as recent information or domain-specific knowledge. This limits their use and trustworthiness, particularly in domains like finance. Retrieval Augmented Generation (RAG) offers a promising solution to improve the factual accuracy of generated answers in such cases, by retrieving relevant information from a database of documents and feeding it into the LLM's context.

RAG on financial documents poses unique challenges. Financial documents such as Annual and Quarterly reports consist of unstructured data, including extensive accounting and financial data in both tabular and textual formats, written with domain-specific language. The challenging nature of answering questions on financial documents has made it an active research area in recent times. We realize the aforementioned benefits of RAG for financial question-answering by experimenting with the latest innovations in RAG.

We carefully select RAG methods that address these common challenges and benchmark them. We apply different chunking techniques such as semantic and contextual chunking. We additionally explore graph and tree based RAG that enable agglomerating data distributed across multiple documents. Finally, we use multimodal RAG to better process tabular data.

2 Related Work

Information retrieval from complex documents is an active area of research. However, little work has been done on financial documents in this regard. LLMs perform poorly when prompted about specific information often contained in financial reports [2, 3]. Financial question-answering benchmarks exist such as the FinQA [2] dataset, which consists of a curated set of financial reports and corresponding questions with ground-truth labels, and FinanceBench [3], which offers a comprehensive financial question-answering benchmark.

Several efforts have been made at fine-tuning LLMs on financial data such as FinGPT [4], BloombergGPT [5] and Instruct-FinGPT [6]. Not only are there issues with access to sufficient high-quality financial data for fine-tuning [7], but it is also impractical to retrain LLMs regularly to update their factual knowledge. Efforts are being made to address the data concern, as in AlphaFin [7], a collection of traditional research datasets, real-time financial data, and handwritten Chain-of-Thought data. However, a retrieval strategy that obviates the need for fine-tuning would not only save costs but provide an efficient way to utilize extensive data from financial reporting without having to continually retrain the LLM.

In a typical RAG system, fetching the most relevant information is the most crucial aspect. In this respect, retrievers have evolved from traditional term-based sparse methods like BM25 [8], TF-IDF [9] to deep-learning-based dense retrieval methods such as DPR [10], ColBERT [11]. Further, Dense Hierarchical Retrieval (DHR) [12], retrieving document candidates first and then relevant passages from the refined document set has been shown to improve performance. Ensemble methods like Hybrid Hierarchical Retrieval (HHR) [13], integrating sparse and dense retrieval methods have also demonstrated performance improvements on the QA datasets. Tree [14] and Knowledge Graph-based methods such as GraphRAG [15] have also been developed to tackle questions directed at longer contexts where the relevant information is scattered across multiple documents or even multiple pages of the same document.

Limited work has been done on using RAG in the finance domain. Unstructured Inc. [16] propose a structural chunking mechanism that improves retrieval on financial reports with unstructured data. They also provide open-source libraries for processing text and tabular data effectively. HybridRAG [17], a combination of GraphRAG and VectorRAG techniques to enhance question-answer (Q&A) systems for information extraction from financial documents has also been recently proposed. But, other state-of-the-art RAG techniques have not extensively been explored in the financial context.

3 Methodology

We use Google’s text-embedding-004 embedding model to embed the documents and queries. We select this model based on its performance on the MTEB benchmark [18] for embedding models. We use Llama 3.1 [19] for generation, except in the multimodal case where we use Gemini 1.5 Flash [20].

Baseline: For our baseline, we use recursive text splitting for chunking, with a chunk size of 1024. After generating the embeddings, they are stored in Chroma DB. Cosine similarity is used to retrieve the top 4 chunks for a query, which are then passed to Llama for generation.

Semantic chunking: Splitting on delimiters can break related text into separate chunks. Semantic chunking addresses this by generating chunks based on the cosine similarity of sentence embeddings. Sentences with cosine similarity above a certain threshold are put in the same chunk.

Fusion Retrieval: Fusion retrieval uses a weighted combination of cosine similarity and BM25, a retrieval algorithm that uses the frequency of keywords, to retrieve the most similar chunks. Semantic chunking is used here as well.

Contextual Retrieval: RAG methods often split documents into smaller chunks, which can result in inefficient retrieval when chunks lack sufficient context. To address this, we implement Contextual Retrieval from Anthropic [21], which enhances each chunk by prepending chunk-specific explanatory context before embedding. Using Gemini 1.5 Flash (Context Size = 1 Mn), we generate this context by prompting with the full document and the respective chunk. To optimize API usage, we batch (up to 50) chunks together using custom prompts. The contextualized chunks are then embedded for downstream generation.

GraphRAG: GraphRAG [15] constructs knowledge graphs from documents to answer queries through indexing and querying phases. In finance, entities might include company names, subsidiaries and key personnel. A graph-based RAG model could tackle this problem by preserving this entity relationship structure inherent in the documents. We created separate graphs for each company for computational feasibility reasons. We also tested a compression-based retriever which uses an LLM to compare the query to the retrieved documents and keep only the text relevant to the query to construct the graph. However, this led to poorer results, likely because there’s a lot of information in the document that is not directly mentioned in the query but is otherwise relevant to answering the question correctly.

Recursive Abstractive Processing for Tree-Organized Retrieval (RAPTOR): We also implemented a tree-based retrieval approach, RAPTOR [14]. RAPTOR recursively clusters chunks of text based on their vector embeddings and generates text summaries of those clusters. The model then uses these summaries to further cluster them, thereby constructing a tree from the bottom up. This tree structure should help account for the semantic depth and hierarchies that exist in the text.

Multi-modal RAG: Here, we use the ColPali technique which uses a visual retriever and ColBERT strategy. We observed that financial reports always contain many crucial information in tables and figures. Hence, in this approach, we treat the PDF as images, replacing the traditional and complex OCR and layout-detection pipelines. Instead, we pass the image patches to a vision language model, PaliGemma-3B, to construct efficient multi-vector embeddings in the visual space. We get a single model that can take into account both the textual and visual content (layout, charts, ...) of a document. At generation time, we pass raw images, tables and text chunks to the multimodal LLM (Google Gemini 1.5 Flash in our experiments) for answer generation.

4 Experiments

4.1 Data

We use the FinanceBench dataset [3]. This dataset consists of 10,231 questions, answers, and evidence triplets. It covers 40 companies that are publicly traded in the United States and 361 public filings, released between 2015 and 2023, including 10Ks, 10Qs, 8Ks, and Earnings Reports.

The questions were developed by annotators with a finance background. The team also created a human evaluation dataset of 150 cases, to compare the model’s performance with human performance. They can be broadly divided into three categories:

- **Domain-relevant:** these are questions that are generically relevant to the financial analysis of companies (such as whether it has paid a dividend in the last year)
- **Metrics-generated:** these are questions that require numerical reasoning and calculations related to financial metrics (such as the asset turnover ratio of the firm)
- **Novel-generated:** these refer to questions that were added by annotators to be more challenging and involve deeper financial reasoning (such as whether a company’s business is subject to cyclicity).

4.2 Evaluation method

The model answers for each of the 150 questions are manually evaluated against the gold answers provided in the dataset, and for each example, the following three labels are assigned: (1) Correct answer: The model answers the question correctly barring minor deviations such as giving the answer in billions when the unit was given in the question as millions. (2) Incorrect answer: The model answers the question incorrectly or gives the correct answer but with the wrong logic or calculations. (3) Refusal to answer: The model refuses to answer the question. Accuracy (percentage of questions correctly answered) is used as the primary evaluation metric to compare between the methods.

5 Results

Our VectorRAG approach achieves 37% correct rate. This is a strong baseline given that [3] reports an accuracy of 19% on their experiments. Four out of six of our techniques perform better than this

strong baseline, with semantic chunking achieving the best performance of 54% correct rate. The multimodal method performed the worst with a 20% correct rate. Refer Fig 1 for accuracy across methods.

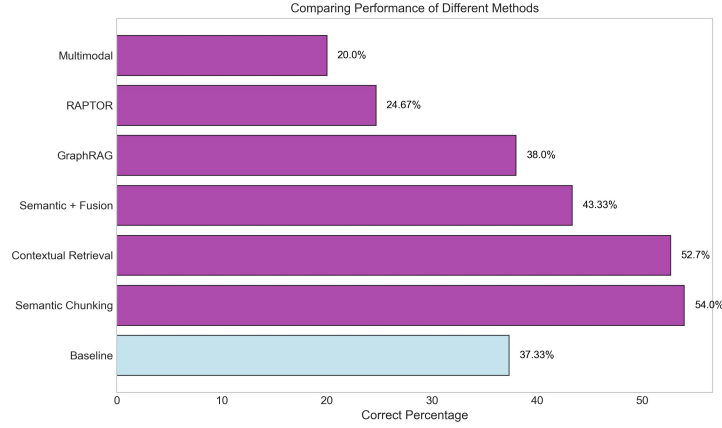


Figure 1: Comparison of % correct across all methods.

A breakdown of each method’s performance by question type is given in Table.1. With this breakdown, each method’s advantages become clearer. Contextual retrieval and semantic chunking excel in all three question types, with semantic chunking being especially superior at Metrics-generated questions (+30% correct rate over baseline)

Contextual retrieval and semantic chunking are the only two methods that refuse less than the baseline across all question types. Both these methods take the surrounding context into account in the chunking process. This is an indication that the additional context helps the model in better retrieval, leading to lesser refusals. On the other hand, Multi-modal and RAPTOR methods refuse drastically more, especially for Metrics-generated questions - 66% and 50% over the baseline refusal rates respectively. This is mostly because the documents retrieved are not relevant to the respective question.

In terms of the incorrect rate, in all but one instance, all the methods perform better than the baseline, with Multi-modal and RAPTOR having the least incorrect across all question types.

6 Conclusions

In this work, we attempt to bridge the gap between LLMs and the demands of finance by building a reliable search tool for finance professionals. We demonstrate that combining state-of-the-art RAG techniques with novel chunking and retrieval methods significantly boosts performance in finance-domain Q&A systems. These systems rely on knowledge bases derived from complex finance documents like 10Ks, 8Ks, and earnings statements. In our experiments, we show that simply applying better chunking techniques (semantic/ contextual) to a Naïve RAG system leads to the best improvements.

7 Future Work

Future work can look at combining these chunking methods with the advanced retrieval techniques we used, to check if that leads to further gain in performance. Furthermore, we have not sufficiently explored prompting techniques like Chain of Thought which could boost performance.

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Appendix

Method	Performance	Domain-relevant	Metrics-generated	Novel-generated
Baseline	% Correct	36%	42%	34%
	% Incorrect	50%	46%	38%
	% Refused	14%	12%	28%
Semantic Chunking	% Correct	48%	72%	42%
	% Incorrect	32%	22%	42%
	% Refused	20%	6%	16%
Contextual Retrieval	% Correct	52%	58%	48%
	% Incorrect	34%	38%	34%
	% Refused	14%	4%	18%
Semantic Chunking + Fusion Retrieval	% Correct	40%	52%	38%
	% Incorrect	40%	32%	34%
	% Refused	20%	16%	28%
GraphRAG	% Correct	34%	42%	38%
	% Incorrect	32%	28%	30%
	% Refused	34%	30%	32%
RAPTOR	% Correct	28%	28%	18%
	% Incorrect	16%	10%	16%
	% Refused	56%	62%	66%
Multi-modal	% Correct	30%	6%	24%
	% Incorrect	28%	16%	34%
	% Refused	42%	78%	42%

Table 1: Collated results from all our methods divided by question type. All percentages reflect accuracy on the test set